MATH 111 Calculus 1, NUU, FALL 2026

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The derivative at a point (Section 2.1, Lecture 7): secant and tangent slopes, the derivative from its definition, tangent lines, average and instantaneous velocity.

Key facts to have at hand (Lecture 7)

- **Secant slope = average rate of change** of f between a and $a+h$: $\dfrac{\Delta y}{\Delta x} = \dfrac{f(a+h)-f(a)}{h}$, the *difference quotient*. It depends on h.
- **The derivative of f at a** is the limit of the secant slopes, if this limit exists:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}.$$

  The two spellings are one number ($x = a+h$). Other names: $\frac{dy}{dx}\big|_{x=a}$ (Leibniz), $\frac{df}{dx}(a)$. (Lecture 7 writes x_0 for a, and Δx or h for the step.)
- **The tangent line** to $y = f(x)$ at $\big(a, f(a)\big)$ is the line through that point with slope $f'(a)$: $y - f(a) = f'(a)(x-a)$. It need not meet the curve only once.
- **The method** (Lecture 7's four steps). (1) Write $f(a+h)$ and open the brackets; (2) subtract $f(a)$ — for fractions, over a common denominator; (3) divide by h — allowed, because the limit never looks at $h = 0$ — and, for roots, multiply by the conjugate, until the h in the denominator cancels; (4) let $h \to 0$. Putting $h = 0$ into the quotient before the h has cancelled gives $0/0$ — an instruction to rewrite, not an answer.
- **Velocity.** If $s(t)$ is the position at time t, then $\dfrac{s(a+h)-s(a)}{h}$ is the *average* velocity over the time interval from a to $a+h$ (a secant slope), and $v(a) = s'(a)$ is the *instantaneous* velocity at time a (a tangent slope). The sign is the direction; the speed is $|v(a)|$.
- **Rates of change and units.** $f'(a)$ is measured in *units of f per unit of x*. Read aloud, $f'(a) = k$ says: *near a, one more unit of x changes f by about k units* — so $f(a+\Delta x) \approx f(a) + k\,\Delta x$ for small Δx.

Warm-up · True or false? (justify with one sentence or one counterexample)

1. A tangent line to a curve meets the curve at exactly one point.
2. $\dfrac{f(3)-f(1)}{2}$ is the slope of the secant line through $\big(1, f(1)\big)$ and $\big(3, f(3)\big)$.
3. $\lim\limits_{h\to 0} \dfrac{f(5+h)-f(5)}{h}$ and $\lim\limits_{x\to 5} \dfrac{f(x)-f(5)}{x-5}$ can be two different numbers.
4. If $f'(2) = 0$, then the tangent line to $y = f(x)$ at $x = 2$ is horizontal.
5. If $s(t)$ is the position (in metres) at time t (in seconds), then $\dfrac{s(5)-s(2)}{3}$ is the velocity at $t = 5$.
6. If $C(x)$ is the cost in dollars of producing x kilograms, then $C'(100)$ is measured in dollars.
7. If $f(3) = 0$, then $f'(3) = 0$.
8. If $f(x) = 7x - 4$, then $f'(a) = 7$ for every number a.

Solution.

1 False. "Touches once" is the circle's tangent from school. The tangent to $y = x^3$ at $(1,1)$ is $y = 3x-2$ (its slope is 3 — Challenge 1 computes it), and $x^3 = 3x-2$ has the second solution $x = -2$: the line meets the curve again at $(-2,-8)$. Lecture 7's warm-up had another: the tangent $y = 1$ to $y = \cos x$ at $x = 0$ touches the curve again every 2π. (Its other picture, $y = x^3$ at 0, showed something else: a tangent may *cross* the curve.) What defines the tangent is *direction* — the limit of secant slopes — not the number of meeting points.

2 True. The slope of a line through two points is rise over run: $\dfrac{f(3)-f(1)}{3-1} = \dfrac{f(3)-f(1)}{2}$. It is also the *average rate of change* of f on $[1,3]$.

3 False. They are the same limit. Substitute $x = 5+h$: then $x \to 5$ exactly when $h \to 0$, and $\dfrac{f(x)-f(5)}{x-5}$ becomes $\dfrac{f(5+h)-f(5)}{h}$. Both are $f'(5)$ (if it exists).

4 True. The tangent at $x = 2$ is $y - f(2) = f'(2)(x-2) = 0$, that is $y = f(2)$: a horizontal line.

5 False. $\dfrac{s(5)-s(2)}{5-2}$ is the *average* velocity over the three seconds from $t = 2$ to $t = 5$ — a secant slope. The velocity *at* $t = 5$ is the limit of such averages over shorter and shorter intervals: $v(5) = s'(5)$. The two agree only by accident.

6 False. A derivative is a rate: $C'(100) = \lim \dfrac{\Delta C}{\Delta x}$ is dollars divided by kilograms, so it is measured in *dollars per kilogram*. (Near 100 kg, one more kilogram costs about $C'(100)$ dollars.)

7 False. The value of f at 3 and the slope of f at 3 are different numbers. Take $f(x) = 2x - 6$: $f(3) = 0$, but every difference quotient is $\dfrac{2(3+h)-6-0}{h} = 2$, so $f'(3) = 2$.

8 True. $\dfrac{f(a+h)-f(a)}{h} = \dfrac{7(a+h)-4-(7a-4)}{h} = \dfrac{7h}{h} = 7$ for every $h \neq 0$, so the limit is 7 whatever a is. A straight line is its own tangent line.

Problem 1 *The derivative from its definition, and the tangent line.*

Find $f'(a)$ from the definition, writing out the four steps; then find an equation of the tangent line to $y = f(x)$ at the point $\big(a, f(a)\big)$.

(a) $f(x) = 3x^2 - 5x,\ \ a = 2$

(b) $f(x) = x^3 - 2x,\ \ a = 1$

(c) $f(x) = \dfrac{2}{x+1},\ \ a = 1$

(d) $f(x) = \sqrt{2x+1},\ \ a = 12$

(e) $f(x) = \dfrac{x}{x-2},\ \ a = 3$

Solution. In every part the same four steps: **(1)** write $f(a+h)$ and open the brackets; **(2)** subtract $f(a)$ (for fractions, over a common denominator); **(3)** divide by h, which is allowed because the limit never uses $h = 0$, and rewrite (for roots, the conjugate) until the h in the denominator cancels; **(4)** let $h \to 0$. Then the tangent line is $y - f(a) = f'(a)(x-a)$.

(a) (1) $f(2+h) = 3(2+h)^2 - 5(2+h) = 3(4+4h+h^2) - 10 - 5h = 2 + 7h + 3h^2$. (2) $f(2) = 12 - 10 = 2$, so $f(2+h) - f(2) = 7h + 3h^2$. (3) $\dfrac{7h+3h^2}{h} = 7 + 3h$ (for $h \neq 0$). (4) $f'(2) = \lim_{h\to 0}(7+3h) = \boxed{7}$. Tangent: $y - 2 = 7(x-2)$, that is $\boxed{y = 7x - 12}$.

(b) (1) $(1+h)^3 = 1 + 3h + 3h^2 + h^3$, so $f(1+h) = 1 + 3h + 3h^2 + h^3 - 2 - 2h = -1 + h + 3h^2 + h^3$. (2) $f(1) = 1 - 2 = -1$, so $f(1+h) - f(1) = h + 3h^2 + h^3$. (3) $\dfrac{h+3h^2+h^3}{h} = 1 + 3h + h^2$. (4) $f'(1) = \lim_{h\to 0}(1 + 3h + h^2) = \boxed{1}$. Tangent: $y - (-1) = 1 \cdot (x-1)$, that is $\boxed{y = x - 2}$.

(c) (1) $f(1+h) = \dfrac{2}{(1+h)+1} = \dfrac{2}{2+h}$. (2) $f(1) = \dfrac{2}{2} = 1$, and over a common denominator $f(1+h) - f(1) = \dfrac{2}{2+h} - \dfrac{2+h}{2+h} = \dfrac{-h}{2+h}$. (3) $\dfrac{1}{h} \cdot \dfrac{-h}{2+h} = \dfrac{-1}{2+h}$ — the h has cancelled. (4) $f'(1) = \lim_{h\to 0} \dfrac{-1}{2+h} = \boxed{-\dfrac{1}{2}}$. Tangent: $y - 1 = -\dfrac{1}{2}(x-1)$, that is $\boxed{y = -\tfrac{1}{2}x + \tfrac{3}{2}}$. (A negative slope: at $x = 1$ the curve is going down, and so is its tangent.)

(d) (1)–(2) $f(12) = \sqrt{25} = 5$ and $f(12+h) = \sqrt{24+2h+1} = \sqrt{25+2h}$, so the difference quotient is $\dfrac{\sqrt{25+2h}-5}{h}$, and $h = 0$ gives $0/0$. (3) A root: multiply by the conjugate $\sqrt{25+2h}+5$ (positive,

so nothing is lost): $\dfrac{\sqrt{25+2h}-5}{h}\cdot\dfrac{\sqrt{25+2h}+5}{\sqrt{25+2h}+5}=\dfrac{(25+2h)-25}{h\big(\sqrt{25+2h}+5\big)}=\dfrac{2}{\sqrt{25+2h}+5}$. (4) $f'(12)=\dfrac{2}{5+5}=\boxed{\dfrac{1}{5}}$. Tangent: $y-5=\dfrac{1}{5}(x-12)$, that is $\boxed{y=\frac{1}{5}x+\frac{13}{5}}$.

(e) (1) $f(3+h)=\dfrac{3+h}{(3+h)-2}=\dfrac{3+h}{1+h}$. (2) $f(3)=\dfrac{3}{1}=3$, and $f(3+h)-f(3)=\dfrac{3+h}{1+h}-\dfrac{3(1+h)}{1+h}=\dfrac{3+h-3-3h}{1+h}=\dfrac{-2h}{1+h}$. (3) Dividing by h: $\dfrac{-2}{1+h}$. (4) $f'(3)=\boxed{-2}$. Tangent: $y-3=-2(x-3)$, that is $\boxed{y=-2x+9}$.

Problem 2 *Average and instantaneous velocity.*

A ball is thrown upwards from the edge of a balcony; it rises, then falls past the balcony to the ground. Its height above the ground t seconds after the throw is

$$s(t)=20+15t-5t^2 \text{ metres} \qquad \text{(until it lands)}.$$

(a) Find the average velocity of the ball over the time intervals $[1,2]$, $[1,1.5]$, $[1,1.1]$ and $[1,1.01]$. What number do these averages approach?

(b) Find the instantaneous velocity at $t=1$ from the definition. Is the ball going up or down?

(c) Find the velocity at an arbitrary time $t=a$. When is the ball at its highest point (where it is momentarily at rest), and how high is it then?

(d) When does the ball hit the ground, and how fast is it going at that instant?

(e) Find the average velocity over the whole flight, from the throw to the landing, and compare it with the velocity at impact.

Solution. It pays to do the algebra once, for a general step h. We have $s(1)=20+15-5=30$ and

$$s(1+h)=20+15(1+h)-5(1+2h+h^2)=30+5h-5h^2,$$

so the average velocity over the time interval from 1 to $1+h$ is

$$\frac{s(1+h)-s(1)}{h}=\frac{5h-5h^2}{h}=5-5h \qquad (h\neq 0).$$

(a) With $h=1,\ 0.5,\ 0.1,\ 0.01$: $5-5h=0,\ \ 2.5,\ \ 4.5,\ \ 4.95$ m/s. (Check the first directly: $s(2)=20+30-20=30=s(1)$, so over $[1,2]$ the ball goes up and comes back to the same height: average velocity 0.) The averages approach 5 m/s.

(b) $v(1)=\lim\limits_{h\to 0}\dfrac{s(1+h)-s(1)}{h}=\lim\limits_{h\to 0}(5-5h)=\boxed{5\text{ m/s}}$. It is positive: the height is increasing, so the ball is still going *up*. The table in (a) predicted the number; the limit proves it.

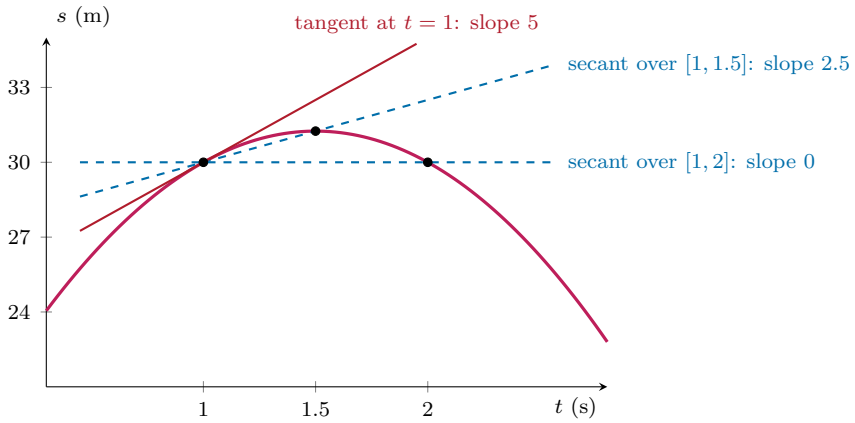
(c) The same computation at an arbitrary time a:

$$s(a+h)-s(a)=15h-5\big((a+h)^2-a^2\big)=15h-10ah-5h^2, \qquad \frac{s(a+h)-s(a)}{h}=15-10a-5h,$$

so $v(a)=15-10a$ m/s. At the highest point the ball is momentarily at rest: $v(a)=0$, i.e. $a=\boxed{1.5\text{ s}}$. The height then is $s(1.5)=20+22.5-11.25=\boxed{31.25\text{ m}}$. (A check with school algebra: completing the square, $s(t)=31.25-5(t-1.5)^2$, which is largest exactly at $t=1.5$. On the graph of s the tangent there is horizontal.)

(d) The ball hits the ground when $s(t)=0$: $20+15t-5t^2=0 \iff t^2-3t-4=0 \iff (t-4)(t+1)=0$. The throw is at $t=0$, so only $t=4$ counts. Then $v(4)=15-40=-25$ m/s: at impact the ball is moving *downwards* at a speed of $\boxed{25\text{ m/s}}$.

(e) Over the whole flight, from $t = 0$ to $t = 4$, the average velocity is $\dfrac{s(4) - s(0)}{4 - 0} = \dfrac{0 - 20}{4} = -5$ m/s: all the average "knows" is that the ball ended 20 m lower than it started, 4 s later. At impact it is moving five times as fast. An average velocity is a secant slope; a velocity at an instant is a tangent slope, and they are different numbers.



The secants from $t = 1$ (dashed) turn towards the tangent (solid) as the interval shrinks.

Problem 3 *Tangent lines with a condition.*

Let $f(x) = x^2 - 4x + 5$.

(a) Show from the definition that $f'(a) = 2a - 4$ for every number a. At which point of the graph is the tangent line horizontal?

(b) Using the formula of (a), find the point at which the tangent line is parallel to the line $y = 2x + 1$, and write the equation of that tangent.

(c) Find every tangent line to the graph that passes through the point $(0, 1)$. (The point is not on the graph.)

Solution.

(a) Fix an arbitrary number a (it stays fixed while $h \to 0$):

$$f(a+h) - f(a) = \big[(a+h)^2 - 4(a+h) + 5\big] - \big[a^2 - 4a + 5\big] = 2ah + h^2 - 4h,$$

so $\dfrac{f(a+h) - f(a)}{h} = 2a + h - 4$ and $f'(a) = \lim_{h\to 0}(2a - 4 + h) = \boxed{2a - 4}$, as claimed. A line is horizontal when its slope is 0: $2a - 4 = 0$, $a = 2$. The point is $\big(2, f(2)\big) = \boxed{(2, 1)}$ and the tangent there is $y = 1$ — the lowest point of the parabola $y = (x-2)^2 + 1$, as expected.

(b) Two lines are parallel when they have the same slope; $y = 2x + 1$ has slope 2. So we need $f'(a) = 2$: $2a - 4 = 2$, $a = 3$. The point is $(3, f(3)) = \boxed{(3, 2)}$, and the tangent is $y - 2 = 2(x - 3)$, that is $\boxed{y = 2x - 4}$ — parallel to $y = 2x + 1$, as required (same slope, different intercept).

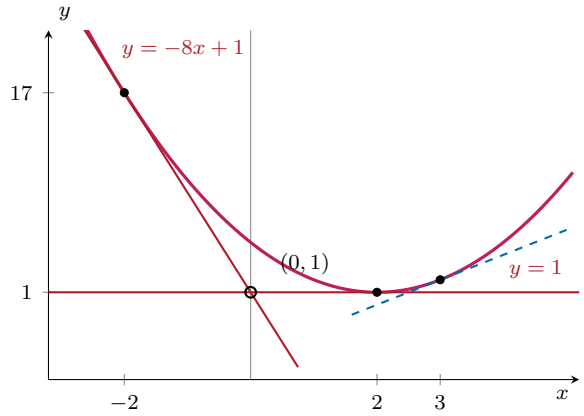
(c) The tangent at the point with first coordinate a is

$$y = f(a) + f'(a)(x - a) = a^2 - 4a + 5 + (2a - 4)(x - a).$$

It passes through $(0, 1)$ exactly when $x = 0$ gives $y = 1$. At $x = 0$: $y = a^2 - 4a + 5 - 2a^2 + 4a = 5 - a^2$. So we need $5 - a^2 = 1$, i.e. $a^2 = 4$, $a = 2$ or $a = -2$. Two tangents:

- $a = 2$: the point $(2, 1)$, slope 0: $\boxed{y = 1}$ (the horizontal tangent of (a));
- $a = -2$: the point $(-2, 17)$, slope $f'(-2) = -8$: $y - 17 = -8(x + 2)$, that is $\boxed{y = -8x + 1}$.

Both indeed pass through $(0, 1)$. (From a point below a parabola that opens upwards there are always two tangents, one on each side.)



The two tangents through $(0,1)$ (solid) and the tangent parallel to $y = 2x + 1$ (dashed).

Problem 4 *Which function, which point?*

Each limit is the derivative $f'(a)$ of some function f at some number a. Say which f and which a, and find the value of the limit.

(a) $\lim_{h\to 0} \dfrac{(3+h)^2 - 9}{h}$ (b) $\lim_{x\to 4} \dfrac{\sqrt{x} - 2}{x - 4}$ (c) $\lim_{h\to 0} \dfrac{\frac{1}{2+h} - \frac{1}{2}}{h}$ (d) $\lim_{x\to 1} \dfrac{x^4 - 1}{x - 1}$

(e) Find a second pair (f, a) that answers (a).

Solution. Compare each limit with the two spellings $\lim_{h\to 0} \dfrac{f(a+h) - f(a)}{h}$ and $\lim_{x\to a} \dfrac{f(x) - f(a)}{x - a}$.

(a) The numerator is $f(3+h) - f(3)$ with $f(x) = x^2$ (indeed $f(3) = 9$): so $f(x) = x^2$, $a = 3$. Value: $\dfrac{9 + 6h + h^2 - 9}{h} = 6 + h \to \boxed{6}$.

(b) This is the $x \to a$ spelling with $f(x) = \sqrt{x}$ and $a = 4$ ($f(4) = 2$). Value: for $x \geq 0$, $x \neq 4$, $\dfrac{\sqrt{x} - 2}{x - 4} = \dfrac{\sqrt{x} - 2}{(\sqrt{x} - 2)(\sqrt{x} + 2)} = \dfrac{1}{\sqrt{x} + 2} \to \boxed{\dfrac{1}{4}}$.

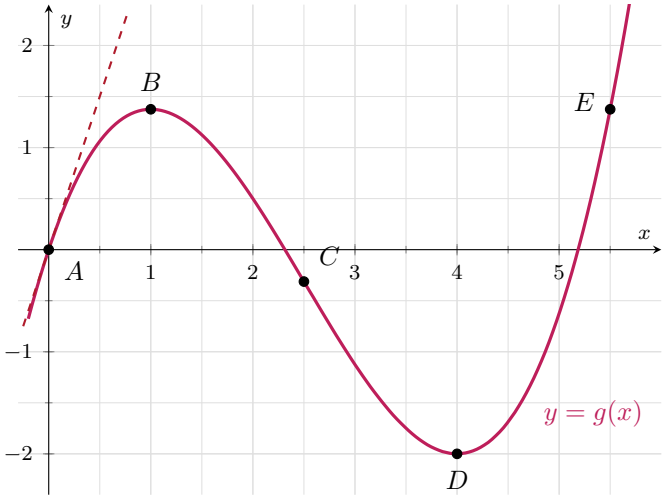
(c) $f(x) = \dfrac{1}{x}$ and $a = 2$. Lecture 7's Example 1 found $f'(x_0) = -\dfrac{1}{x_0^2}$ for this f, so the limit is $\boxed{-\dfrac{1}{4}}$. Directly: $\dfrac{1}{2+h} - \dfrac{1}{2} = \dfrac{-h}{2(2+h)}$, and dividing by h gives $\dfrac{-1}{2(2+h)} \to -\dfrac{1}{4}$.

(d) $f(x) = x^4$ and $a = 1$ ($f(1) = 1$). Value: $x^4 - 1 = (x-1)(x^3 + x^2 + x + 1)$, so for $x \neq 1$ the quotient is $x^3 + x^2 + x + 1 \to \boxed{4}$.

(e) The answer to (a) is not unique. With $f(x) = (3+x)^2$ and $a = 0$: $\dfrac{f(0+h) - f(0)}{h} = \dfrac{(3+h)^2 - 9}{h}$, the same limit. Or $f(x) = x^2 + 7$ and $a = 3$: the constant 7 cancels in $f(3+h) - f(3)$.

Problem 5 *Reading slopes off a graph.*

The graph of a function g is shown, with five marked points A–E; the dashed line is the tangent to the graph at A.



(a) At which of the points A–E is the derivative positive? negative? zero?

(b) Use the dashed tangent line to find $g'(0)$.

(c) Arrange in increasing order: 0, $g'(0)$, $g'(2.5)$, $g'(5.5)$, and the slope of the secant line through A and E.

(d) (A different function.) The tangent line to $y = k(x)$ at the point $(2, 5)$ passes through $(6, -3)$. Find $k(2)$ and $k'(2)$.

Solution. The derivative at a point is the slope of the tangent there; so read slopes, not heights.

(a) Draw the tangent at each point, as the dashed one is drawn at A. At A and at E the tangent goes up to the right: $g' > 0$. At C it goes down to the right: $g' < 0$. At B (the top of the hump) and at D (the bottom of the dip) the tangent is horizontal: $g' = 0$. (Note that B is above the x-axis and D below it, yet the slope is 0 at both, while at A the value is 0 and the slope is 3: the sign of the *value* says nothing about the *slope*.)

(b) The dashed tangent at A passes through $(0, 0)$ and through the grid point $(0.5, 1.5)$, so its slope is $\dfrac{1.5 - 0}{0.5 - 0} = 3$: $g'(0) = 3$.

(c) $g'(2.5)$ is negative (the tangent at C goes down to the right). The secant through $A(0, 0)$ and E, which is at about $(5.5,\ 1.4)$ on the picture, rises gently: its slope is about $\dfrac{1.4}{5.5} \approx 0.25$, positive but small. At A the tangent has slope 3, and the tangent at E is steeper still. So

$$g'(2.5) \ < \ 0 \ < \ \text{slope of } AE \ < \ g'(0) \ < \ g'(5.5).$$

(For the record, the function is $g(x) = \frac{1}{4}x^3 - \frac{15}{8}x^2 + 3x$, and the five numbers are -1.6875, 0, 0.25, 3 and 5.0625.)

(d) The tangent line touches the graph at $(2, 5)$, so $k(2) = \boxed{5}$, and its slope is the derivative: $k'(2) = \dfrac{-3 - 5}{6 - 2} = \dfrac{-8}{4} = \boxed{-2}$.

Challenge *(for those who finish early, or at home)*

1. Find, from the definition, the slope of the tangent to $y = x^3$ at $(1, 1)$, and the equation of the tangent. Show that this tangent meets the curve at a second point, and find it.
2. Take $x_0 > 0$ and the point $P = \left(x_0, \frac{9}{x_0}\right)$ on the hyperbola $y = \frac{9}{x}$. Show that the tangent at P cuts off from the first quadrant a triangle of area 18, wherever P is, and that P is the midpoint of the part of the tangent that lies between the axes. (Lecture 7 did this for $y = \frac{1}{x}$.)

Solution.

1. $\dfrac{(1+h)^3-1}{h} = \dfrac{3h+3h^2+h^3}{h} = 3+3h+h^2 \to 3$, so the slope is 3 and the tangent is $y-1=3(x-1)$, that is $y=3x-2$. It meets the curve where $x^3=3x-2$, i.e. $x^3-3x+2=0$. We know $x=1$ is a root, and it is a double one: $x^3-3x+2=(x-1)^2(x+2)$. So the tangent meets the curve again at $x=-2$, the point $(-2,-8)$ — a tangent line need not meet the curve only once (warm-up, item 1).

2. *The slope,* as in Lecture 7's Example 1: $\dfrac{1}{h}\Big(\dfrac{9}{x_0+h}-\dfrac{9}{x_0}\Big) = \dfrac{1}{h}\cdot\dfrac{9x_0-9(x_0+h)}{x_0(x_0+h)} = \dfrac{-9}{x_0(x_0+h)} \longrightarrow -\dfrac{9}{x_0^2}$. *The tangent:* $y-\dfrac{9}{x_0} = -\dfrac{9}{x_0^2}(x-x_0)$. *The intercepts.* Put $y=0$: $-\dfrac{9}{x_0} = -\dfrac{9}{x_0^2}(x-x_0)$, so $x-x_0=x_0$ and $x=2x_0$. Put $x=0$: $y=\dfrac{9}{x_0}+\dfrac{9}{x_0}=\dfrac{18}{x_0}$. *The area* of the triangle with legs $2x_0$ and $\dfrac{18}{x_0}$ is $\dfrac{1}{2}\cdot 2x_0\cdot\dfrac{18}{x_0}=18$ — the x_0 cancels, so the area does not depend on where P is. *The midpoint* of $(2x_0,0)$ and $\big(0,\frac{18}{x_0}\big)$ is $\big(x_0,\frac{9}{x_0}\big)$, which is P. (For $y=\frac{c}{x}$ the same computation gives area $2c$; Lecture 7's $y=\frac{1}{x}$ is $c=1$, area 2.)